

Internship Rapport

Program:

- Introduction of the systems used by the bank
- Networks used by the bank
- Network Services and Systems
- Security and Monitoring Systems
- Network Automation and Programmability
- Practice and Workshops

1/ Introduction of the systems used by the banks

-The bank is divided into core systems, and support systems, core systems are the systems at the heart of the business goals for the bank it's mainly the core banking, support systems are everything else like the networking and Jira system

-the bank mainly uses Windows and ADDS, t24, Jira and cisco equipment

2/ Network design by the bank

There are mainly 7 kinds of Networks :

1. Campus Network: its is a LAN network (also knows as CAN) located In the HQ
2. Branches Network (agencies): the network that links the branches to the HQ (mdc & bdc)
3. Production Network(DC-PROD): located in HQ underground, Backup located in Blida that hosts banks services and applications
4. Management Network for managing different IT platforms (network security, infra-system, support)
5. Swift Network: for international banking transactions
6. Partners Network: for partners interconnection (BA,CNIS,CPI,SATIM,...etc)
7. Hosting Network / DMZ: for hosting applications / services that publicly (from internet) accessible by external users

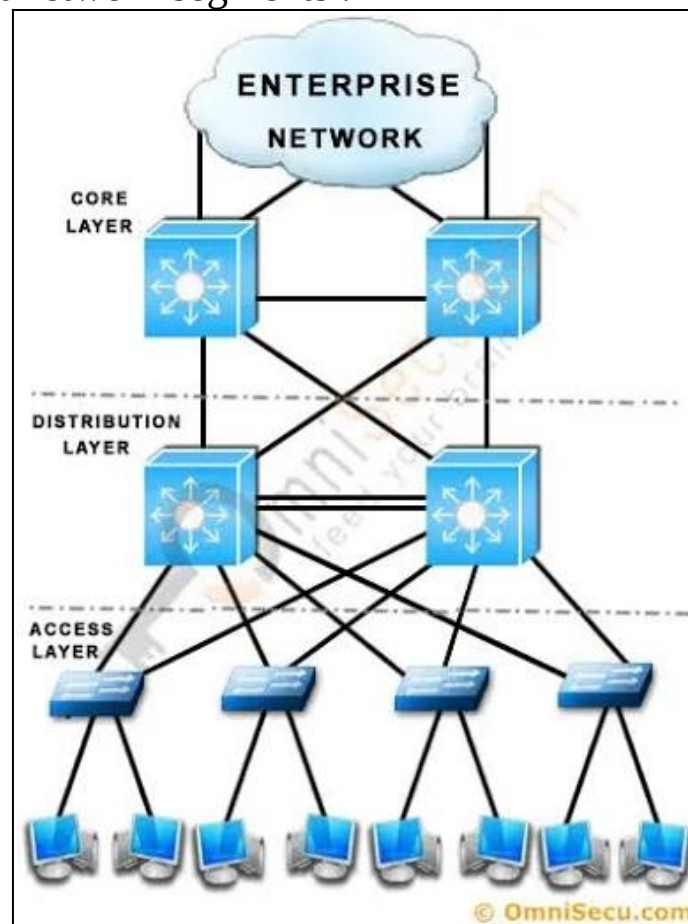
3/ Network Services and Systems

~~-The Bank network Architecture is based on the departments as each department and sub-department has its own network~~

- The design of the campus network is based on three-tier cisco hierarchical network design (As illustrated on the figure below)

-**the access layer** contains access switches distributed across the HQ, annexes and branches which are ~~possible~~ responsible for connecting bank's users, voice-IP (IP telephony), other peripherals printers, cameras, ATM machines ...etc.

-**The distribution and the core layer** are collapsed into a single layer, which contains stacked **multi-layer switch (MLS)** that act as the **gateway** for all the networks **VLANs** (used by the campus) , it ensures the routing between different network segments .



-the campus consists a main site and 5 annexes listed below:

- 1st annex (IT Annex) has 5 access switches including one of them designed as master that links to the core via fiber connection
- 2nd annex (HR Annex) has 5 access switches including one of them designed as master that links to the core via fiber connection
- 3rd annex (CIDAR Annex) has 4 access switches including one of them designed as master that links to the core via fiber connection
- 4th annex (Blanche Annex) has 3 access switches including one of them designed as master that links to the core via fiber connection
- 5th annex (RETAIL Annex) has 4 access switches, including one of them designed as master that links to the core via fiber connection

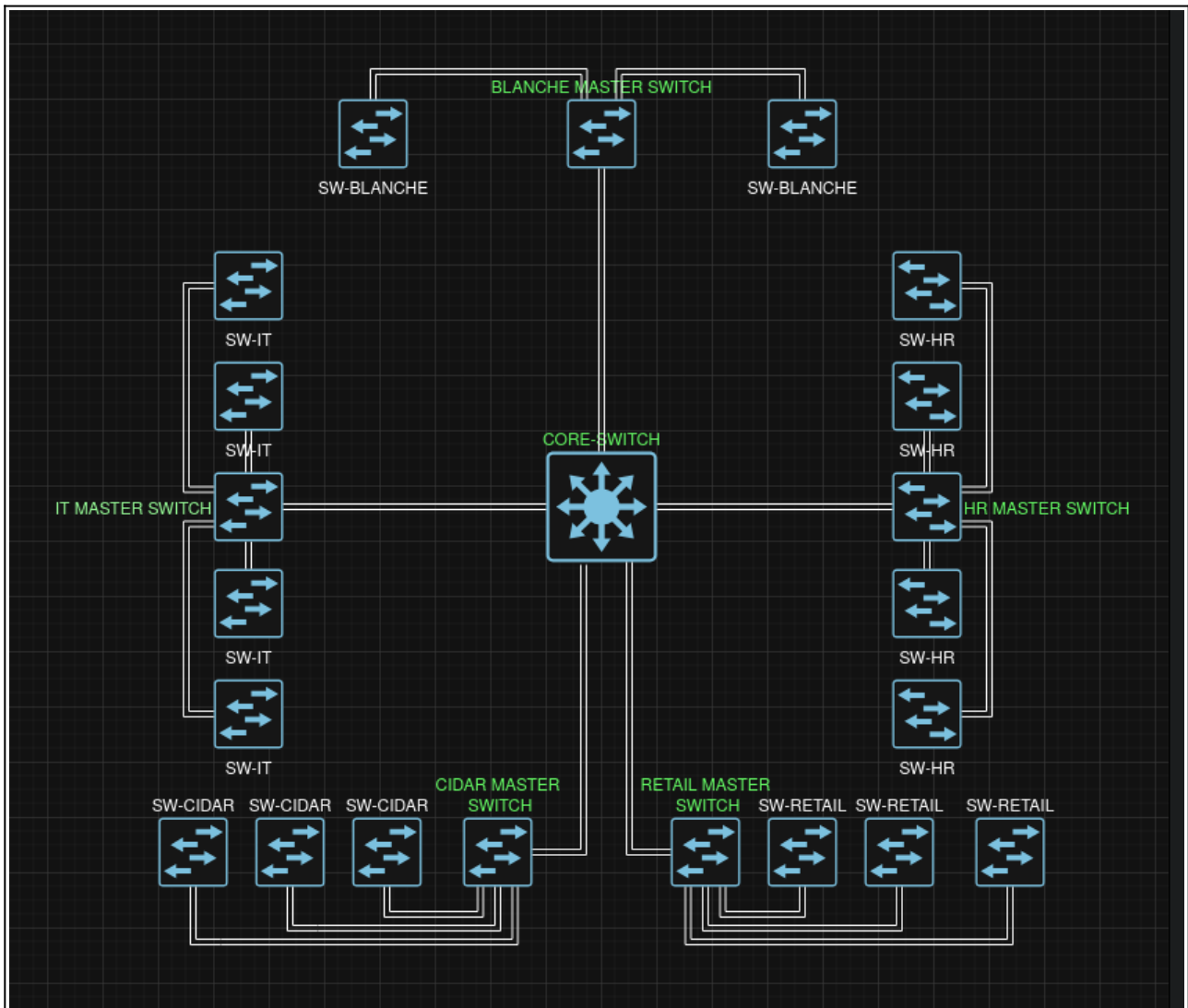
~~-Traffic to the configuration VLAN is ran through a core switch that acts as the default gateway for communication~~

- the vlans segmentation design is based on the hierarchy of the bank's structure, there are three types of vlans, data, voice and management

- data: carry out user traffic
- voice: carry out voice IP traffic
- management: for managing equipment

there are two types of interfaces in a switches (VLAN), Access and Trunk:

- Access for endpoints (pc, phone, printer...etc)
- Trunk for switch to router, switch to switch or switch to firewall



-access to switches and other equipment and services is only allowed using jump servers, a layer between the actual switch and the desktop, accessed by remote desktop protocol, sessions are saved for security and practical reasons.

-access to jump servers is given by the DSSI department via ADDS groups.

-Both **beyond trust privileged remote access** and **mobaxterm** (ssh) is used in throughout the process

-the switches all have a special configuration VLAN configured as a loop-back interface with special access privileges.

-the Switches and Routers use a mix of a firewall and ACL to configure trusted hosts, the trusted hosts are the jump servers, only they are allowed to access the equipment.

-some useful info about switches, VLAN configuration is saved in .dot file in flash drive

TODO:

remake graph with draw.io

- look into switch and vlan operation

vlan {id}

name {name}

interface:

 switchport mode access

 switchport mode vlan {id}

show vlan brief

l load

- read up more of hierchical networks

hierarchical network architecture contains three layer, access layer mostly made up of access switches with the purpose of forming multiple LANs, distribution layer above it made of layer 3 switches and router that it handles routing of said LANs, and a core layer at the very top with the soul purpose of being incredibly fast and fault tolerant, when a network is too small it is possible to merge the distribution and the core layers into one.

This model is considered the industry standard as it is efficient, scalable, secure and easy to manage

- eve-ng or pnetlab simulators
downloaded, not yet used
- ether channel (LACP, PAGP)

an ether channel is a logical link that groups multiple physical links at once for a maximum of 8 ports, improving speed and redundancy, since it only one logical link it baypasses stp it can be done with two protocols, PAgP and LACP

PAgP	LACP
Cisco proprietary	IEEE open standard
on/off/desirable/auto	on/off/active/passive

syntace:

1. interface range fastethernet0/1-fastethernet0/3
2. channl-protocol {lacp or pagp}
3. channel-groupe {logical-number} mode {active/desirable}

after that a logical interface called port-channel{logical-number} is created

remaining program:

Services Réseaux : Core network and security services (DNS, DHCP, AD DS + GPO coordination infrasystem) :

Comprendre l'architecture des services + implementation ,

Sécurité : AV , NAC , FW (NGFW), proxy, waf, load balancer , IP telephony , wireless , routing& qwitches , transmission ,

Incentoring , snipeIT, netbox , bitwardne , logging , monitoring , SOC ; -> securityetc , fotianalyzer , fortiamanger

NOC (PRTG, Whatsupp) ,

Network automation & programability ; python , ansible, netmiki

Pratique: démonstrations , simulations, workshop , hands-on

te=ten gig
todo:

- presentation stp
- spine and leaf
- vtp ctp